

Week 9 Game Theory Notes

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1 Continuous Strategies

1.1 From Finite Games to Continuous Strategy Sets

In the finite setting, ordinal potentials preserve each player's preference ordering over unilateral deviations. This idea can be extended to continuous strategy spaces by replacing the payoff matrix pair (Π^1, Π^2) with two payoff functions

$$\Pi^1 \longrightarrow u_1(x, y), \quad \Pi^2 \longrightarrow u_2(x, y),$$

where x and y represent strategy variables rather than probabilities.

A possible example of these payoff functions is

$$u_1(x, y) = -\left(x - \frac{1}{2}\right)^2 - (x - y)^2, \quad u_2(x, y) = -\left(y - \frac{1}{2}\right)^2 - (x - y)^2,$$

with $(x, y) \in [0, 1]^2$.

These two objective functions reward each player for staying close to the midpoint of the interval while also penalizing disagreement between the two strategies. The corresponding Nash equilibrium of this system is at $(\frac{1}{2}, \frac{1}{2})$

1.2 A Continuous Ordinal Potential

Like payoff matrices, ordinal potentials can also be extended to the continuous strategy case. For the example payoff functions above, we can define the ordinal potential

$$\Phi(x, y) = -\left(x - \frac{1}{2}\right)^2 - \left(y - \frac{1}{2}\right)^2 - (x - y)^2.$$

This function preserves each player's directional incentives. In the continuous case, we check this preservation through partial derivatives:

$$\frac{\partial}{\partial x} u_1(x, y) = -2\left(x - \frac{1}{2}\right) - 2(x - y) = \frac{\partial}{\partial x} \Phi(x, y),$$

and

$$\frac{\partial}{\partial y} u_2(x, y) = -2\left(y - \frac{1}{2}\right) + 2(x - y) = \frac{\partial}{\partial y} \Phi(x, y).$$

Proving that these partial derivatives are equal shows that the potential captures Player 1's incentive with respect to x and Player 2's incentive with respect to y . Like the discrete ordinal potential, the continuous ordinal potential preserves each player's unilateral improvement directions.

1.3 Nash Equilibrium via the Potential

A Nash equilibrium (x^*, y^*) occurs when neither player benefits from changing their strategy unilaterally. In this example, the critical-point conditions are

$$\left. \frac{\partial}{\partial x} \Phi(x, y) \right|_{(x^*, y^*)} = 0, \quad \left. \frac{\partial}{\partial y} \Phi(x, y) \right|_{(x^*, y^*)} = 0.$$

Substituting the above expressions yields

$$-2\left(x - \frac{1}{2}\right) - 2(x - y) = 0, \quad -2\left(y - \frac{1}{2}\right) + 2(x - y) = 0.$$

Solving this system gives

$$x^* = \frac{1}{2}, \quad y^* = \frac{1}{2}.$$

Therefore, the example continuous game has Nash equilibrium at

$$(x^*, y^*) = \left(\frac{1}{2}, \frac{1}{2}\right).$$

1.4 Existence Results

For finite two-player games, an ordinal potential $F(i, j)$ must satisfy:

$$F(i, j) > F(k, j) \iff \Pi_{ij}^1 > \Pi_{kj}^1, \quad F(i, j) > F(i, k) \iff \Pi_{ij}^2 > \Pi_{ik}^2.$$

A standard consequence is that if a finite game admits an ordinal potential, then it possesses a pure-strategy Nash equilibrium [Monderer and Shapley, 1996].

For compact continuous strategy sets, the slides state the generalized ordinal potential condition:

$$\Phi(x, y) > \Phi(z, y) \implies u_1(x, y) > u_1(z, y), \quad \Phi(x, y) > \Phi(x, z) \implies u_2(x, y) > u_2(x, z).$$

Under compactness, existence of such a generalized ordinal potential implies existence of a Nash equilibrium [Kukushkin, 2010].

1.5 Relation to Conservative Potentials

We can interpret the ordinal potential through vector calculus. Consider the vector field

$$\mathbf{V}(x, y) = \left(\frac{\partial}{\partial x} u_1(x, y), \frac{\partial}{\partial y} u_2(x, y) \right).$$

An ordinal potential exists precisely when this field is conservative, so that there is a scalar function Φ satisfying

$$\frac{\partial}{\partial x} \Phi(x, y) = \frac{\partial}{\partial x} u_1(x, y), \quad \frac{\partial}{\partial y} \Phi(x, y) = \frac{\partial}{\partial y} u_2(x, y).$$

If the strategy domain is simply connected, a necessary and sufficient condition is the zero-curl condition

$$\frac{\partial^2}{\partial y \partial x} u_1(x, y) = \frac{\partial^2}{\partial x \partial y} u_2(x, y).$$

1.6 Example of the Zero-Curl Test

Consider the example

$$u_1(x, y) = x^2 y - \frac{3}{4} x, \quad u_2(x, y) = x^2 y - \frac{1}{4} y.$$

Then

$$\frac{\partial}{\partial y} \left(\frac{\partial u_1}{\partial x} \right) = \frac{\partial}{\partial y} (2xy - \frac{3}{4}) = 2x, \quad \frac{\partial}{\partial x} \left(\frac{\partial u_2}{\partial y} \right) = \frac{\partial}{\partial x} (x^2 - \frac{1}{4}) = 2x.$$

After collecting terms appropriately, the potential is written as

$$\Phi(x, y) = x^2y - \frac{3}{4}x - \frac{1}{4}y$$

1.7 Discrete Zero-Curl Condition

There exists a discrete analogue of the zero-curl test. The idea is to compare local preference changes around orthogonally adjacent cells.

For a 3×3 bimatrix game, one first computes signed local preference differences for Player 1 along columns and for Player 2 along rows. Then the local preferences are computed again, this time along rows for Player 1 and among columns for Player 2. These local differences are then compared between each orthogonally adjacent square. If the directional preference data in both reduced matrices are compatible around each square, the game behaves like a discrete conservative field and an ordinal potential exists. If this local preference data disagrees between the reduced matrices, then no ordinal potential exists.

2 Stability of Numerically Modeling BNN with RK4

2.1 The RK4 Method

I also studied the Brown–von Neumann–Nash (BNN) dynamical system under fourth-order Runge–Kutta discretization.

For a general ODE

$$\frac{dy}{dt} = f(t, y), \quad y(t_0) = y_0,$$

the classical RK4 method uses the increments

$$\begin{aligned} k_1 &= f(t_n, y_n), \\ k_2 &= f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_1\right), \\ k_3 &= f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_2\right), \\ k_4 &= f(t_n + h, y_n + hk_3), \end{aligned}$$

and updates by

$$y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4), \quad t_{n+1} = t_n + h.$$

2.2 Absolute Stability of RK4

The RK4 stability polynomial is as follows:

$$R(z) = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24},$$

which is the fourth-order Taylor truncation of e^z . For a Jacobian with eigenvalues λ_i and step size h , the local absolute-stability condition is

$$|R(h\lambda_i)| \leq 1 \quad \forall i$$

The numerical stability of the discretized BNN dynamics is thus determined by whether the eigenvalues of the Jacobian (scaled by the step size) fall within the RK4 stability region.

2.3 Simple BNN System: Two Strictly Dominant Strategies

The first example was a two player game with 2×2 payoff matrices, where both players had a strictly dominant strategy.

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 5 & 7 \\ 6 & 8 \end{pmatrix}.$$

where we defined $p^1 = (x^1, y^1)$ as Player 1's probability vector and $p^2 = (x^2, y^2)$ as Player 2's probability vector. But the laws of probability:

$$x^1 + y^1 = 1, \quad x^2 + y^2 = 1.$$

In this game, Player 1's second strategy strictly dominated their first, and Player 2's second strategy strictly dominated their first.

This led to the following φ functions:

$$\varphi_{11}(p) = 0, \quad \varphi_{12}(p) = 2x^1, \quad \varphi_{2A}(p) = 0, \quad \varphi_{2B}(p) = 2x^2.$$

Therefore the BNN system

$$\dot{p}_{ij} = \varphi_{ij}(p) - p_{ij} \sum_k \varphi_{ik}(p)$$

reduces to

$$\begin{aligned} \dot{x}^1 &= -2(x^1)^2, & \dot{y}^1 &= 2(x^1)^2, \\ \dot{x}^2 &= -2(x^2)^2, & \dot{y}^2 &= 2(x^2)^2. \end{aligned}$$

The eigenvalues of the corresponding Jacobian matrix are:

$$\lambda = 0, \quad \lambda = 0, \quad \lambda = -4x^1, \quad \lambda = -4x^2.$$

Since all nonzero eigenvalues are real and nonpositive, the RK4 step-size condition becomes

$$4hx^1 \leq 2.784., \quad |R(-4hx^2)| \leq 1.$$

Because the inferior-strategy probabilities x^1 and x^2 decrease over time, the restriction holds for the initial state.

2.4 A More Complicated BNN System: One Strictly Dominant Strategy

The second example changed the payoff matrices to

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 7 & 5 \\ 6 & 8 \end{pmatrix}.$$

Here Player 1 still has a strictly dominant second strategy, but Player 2 no longer has a strictly dominant strategy.

Now, both of Player 2's φ functions were dependent on x^1 and y^1 while Player 1's φ functions stayed the same:

$$\varphi_{11}(p) = 0, \quad \varphi_{12}(p) = 2x^1,$$

and

$$\begin{aligned} \varphi_{2A}(p) &= \max(0, 7x^1 + 6y^1 - 7x^1x^2 - 5x^1y^2 - 6y^1x^2 - 8y^1y^2) \\ &= \max(0, 2x^1y^2 - 2y^1y^2) = 2y^2 \max(0, x^1 - y^1), \\ \varphi_{2B}(p) &= \max(0, 5x^1 + 8y^1 - 7x^1x^2 - 5x^1y^2 - 6y^1x^2 - 8y^1y^2) \\ &= \max(0, -2x^1x^2 + 2y^1x^2) = 2x^2 \max(0, y^1 - x^1). \end{aligned}$$

The ODE became:

$$\begin{aligned} \dot{x}^1 &= -2(x^1)^2, & \dot{y}^1 &= 2(x^1)^2, \\ \dot{x}^2 &= 2(y^2)^2 \max(0, x^1 - y^1) - 2(x^2)^2 \max(0, y^1 - x^1), \\ \dot{y}^2 &= 2(x^2)^2 \max(0, y^1 - x^1) - 2(y^2)^2 \max(0, x^1 - y^1). \end{aligned}$$

Because of the max terms, the Jacobian must be analyzed case by case.

Case 1: $x^1 > y^1$. In this region,

$$\dot{x}^2 = 2(y^2)^2(x^1 - y^1), \quad \dot{y}^2 = -2(y^2)^2(x^1 - y^1),$$

with Jacobian eigenvalues

$$0, 0, -4x^1, -4y^2(x^1 - y^1).$$

Case 2: $y^1 > x^1$. In this region,

$$\dot{x}^2 = -2(x^2)^2(y^1 - x^1), \quad \dot{y}^2 = 2(x^2)^2(y^1 - x^1),$$

with Jacobian eigenvalues

$$0, 0, -4x^1, -4x^2(y^1 - x^1).$$

As in the previous example, all nonzero eigenvalues are nonpositive in their respective regions.

2.5 An Even More Complicated BNN System: No Strictly Dominant Strategies

In the last example, the following payoffs were used:

$$A = \begin{pmatrix} 2 & 3 \\ 4 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 7 & 5 \\ 6 & 8 \end{pmatrix}.$$

In this case, neither player had a strictly dominant strategy.

The corresponding BNN ODE was now:

$$\begin{aligned} \dot{x}^1 &= 2(y^1)^2 \max(0, y^2 - x^2) - 2(x^1)^2 \max(0, x^2 - y^2), \\ \dot{y}^1 &= 2(x^1)^2 \max(0, x^2 - y^2) - 2(y^1)^2 \max(0, y^2 - x^2), \\ \dot{x}^2 &= 2(y^2)^2 \max(0, x^1 - y^1) - 2(x^2)^2 \max(0, y^1 - x^1), \\ \dot{y}^2 &= 2(x^2)^2 \max(0, y^1 - x^1) - 2(y^2)^2 \max(0, x^1 - y^1). \end{aligned}$$

The Jacobian again depended on the region of state space. For this example, there were four different regions based on the max inputs and the φ functions:

- (i) $x^1 > y^1$ and $x^2 > y^2$,
- (ii) $x^1 > y^1$ and $y^2 > x^2$,
- (iii) $y^1 > x^1$ and $x^2 > y^2$,
- (iv) $y^1 > x^1$ and $y^2 > x^2$.

For each region, the Jacobian simplified to a polynomial matrix in (x^1, y^1, x^2, y^2) , and the eigenvalues depend on the inferior-strategy mass and on the gap between the competing strategy probabilities.

I conjectured that the dominant bound comes from an eigenvalue of magnitude at most about 4, leading to the rough heuristic condition

$$h \lesssim 0.696.$$

That said, the eigenvalue analysis still needs further investigation.

2.6 Numerical Observations

The numerical slides compare RK4 trajectories for several step sizes and several initial conditions. The qualitative conclusions are:

- (1) stronger dominance appears to enlarge the set of stable step sizes;
- (2) once strict dominance weakens or disappears, the admissible RK4 step size becomes more sensitive to initial conditions;
- (3) in the no-dominance case, the dynamics can still converge numerically for some step sizes, but the safe stability region is much less transparent from the local Jacobian calculations alone.

While the BNN flow may remain well behaved in continuous time, a fixed-step RK4 approximation can become unstable when the local Jacobian spectrum approaches the boundary of the RK4 absolute-stability region.